

2) Artificial Intelligence Assignment Q1:
Name: Yash Ray
Roll no: 24K-0737

Question #1.

Selected System: IBM Deep Blue

part(a).

Core objective:-

Primary objective of Deep Blue was to defeat a chess champion under standard tournament conditions.

Theoretical foundation:-

Deep-Blue is based on:

- (i) Game Theory
- (ii) State space search
- (iii) Minimax decision rule
- (iv) Alpha Beta pruning
- (v) Heuristic evaluation functions

Reflection:-

Its design reflects core AI goals:

- (i) Rational decision making.
- (ii) Problem-solving under constraints.
- (iii) Acting optimally in competitive environment.

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part(b)

Input-interface
(Chess Board state)



Move Generator
(Legal moves)



Search Engine
(minimax + AlphaBeta)



Evaluation-function
(Heuristics)



Decision-module
(Best move output)

Role of Each module:-

1. Input-interface:- Receive board configuration.
2. Move-generator:- Generate all legal moves.
3. Search Engine:- Explores possible future states.
4. Evaluation-function:- Assign numerical score to position.
5. Decision-module:- Choose move with highest evaluation.

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part (c).

Key Components:-

- (i) Specialized Hardware:- Parallel processors for massive computation.
- (ii) Search Algorithm:- Minimax + AlphaBeta.
- (iii) Evaluation function:- Encoded Expert-Knowledge.
- (iv) Opening database:- Pre-analyzed input openings.
- (v) Endgame Tablebases:- Pre-computed optimal endings.

2) Interaction of Components:-

- (i) Hardware enabled deep search.
- (ii) Heuristic guided pruning efficiently.
- (iii) Opening & endgame databases improved results.
- (iv) Integration produced near perfect tactical accuracy.

2) Deep Blue demonstrated highly optimized narrow intelligence.

Done

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Part (d):

Impacts on Research, Industry & Society:

1. Research:-

Deep-Blue showed that machines could outperform humans in complex intellectual tasks. Sparking interest in AI-research, it also influenced the development of search algorithms & parallel-computing.

2. Industry:-

Boosted confidence in AI systems.
Influenced finance, logistic optimization.

3. Society:-

Changed perception of human vs machine capability.

Type of Intelligence:-

2) Deep Blue represents Narrow AI as:-

- (i) It can't learn outside chess.
- (ii) It lacks understanding or consciousness.
- (iii) It operates via computation, no human cognition.

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part(e).

Limitations:

(i) Technical:

- No learning capability.
- Domain specific.

(ii) Ethical:

- Misinterpretation of machine-intelligence.
- Raised fear of human replacement.

(iii) Practical:

- Extremely Expensive.
- Not Adaptable.

part(f).

z) future-improvements:

(i) Integrate machine-learning.

(ii) Use reinforcement learning.

(iii) Develop explainable decision-modules.

(iv) Reduce energy.

(v) Reduce hardware dependency.

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Question # 2

- => Selected field:- Healthcare
- => Technology:- Deep-learning for medical Imaging.

part (a) - Emerging AI Technology:-

↳ AI-powered Diagnostic Systems (Deep-learning in radiology)

Ex:- AI detecting cancer from MRI-scans.

part (b) - Working-mechanism:-

- (i) Medical images collected.
- (ii) Convolutional Neural Network trained.
- (iii) Model learns patterns of abnormalities.
- (iv) Output probability of disease.

Applications:-

- (i) Cancer-detection
- (ii) Covid-19 diagnosis
- (iii) Brain tumor detection
- (iv) Pathology slide analysis

Names Yash-Raj

Roll-no: 24k-0734

part(c):-

Advantages:-

- High accuracy.
- Fast diagnosis
- Reduces doctor work-load

Challenges:-

- Requires large labelled datasets
- Data bias
- Lack of interpretability

Ethical Issues:-

- Patient-privacy
- Bias against certain demographics
- Over-reliance on AI

part(d):-

future-development:-

- Explainable AI (XAI)
- Federated learning for privacy
- AI-assisted (not AI-replacing) diagnosis
- Regulatory frameworks

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Question # 31

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part (a) - PEAS Description:

(i) AI-based Resume Screening System:

Performance measure: Accurate candidate ranking.

Environment: Job Applications

Actuators: Shortlisting / rejection

Sensors: Resume Text, Keywords

(ii) Flood Detection AI:

Performance: Early flood warnings.

Environment: Rivers, rainfall zone.

Actuators: Alerts, Alarms

Sensors: Water levels, rainfall sensors.

part (b):

Agent	Observable	Deterministic	Episodic	Dynamic	Discrete
Resume AI	Fully	Deterministic	Episodic	Static	Discrete
Flood AI	Partially	Stochastic	Sequential	Dynamic	Continuous

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Question #41 (Automated library Retrieval system)

- State-space: $(x, y, \text{carrying book?})$
- Initial-state: start-position, No book.
- Actions: Move (up/down/right/left), pick.
- Transition-Model: Position-change.
- Goal-Test: At target shelf with book.
- Path-costs: Time + movement cost.
- Solutions: Shortest valid path.

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Question #1 (part a):

Search strategy: The drone is using "Iterative Deepening Search (IDS)" which is a combination of DFS & BFS.

How the drone explores:

1. It explores all paths upto a fixed depth-limit.
2. If the exit is not found within that depth it returns to the starting position.
3. It increases the depth limit slightly.
4. It repeats the search.
5. This continues until the exit is discovered.

Question #1 (part b): Why this is appropriate?

(a) Limited-memory:

- IDS behave like DFS in memory usage.
- It only stores a single path from root to leaf.

(b) Unknown Depth of the exit:

- ↳ IDS gradually increases depth.
- if exit is shallow → found quickly.
- if exit is deep → Eventually found.

→ IDS is Complete even when solution depth is unknown.

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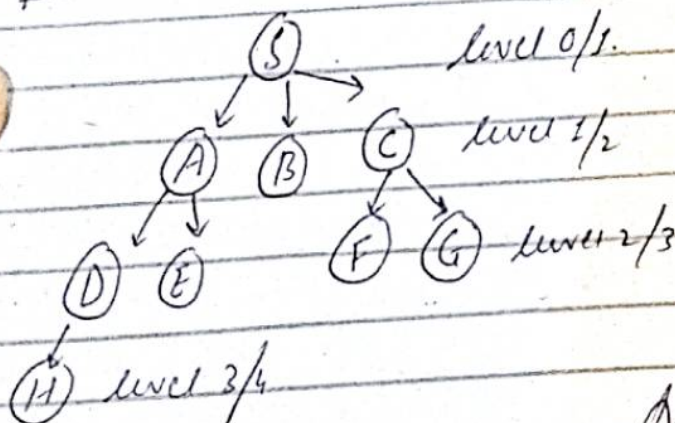
(c) Very Deep dead End Tunnel

IDS prevents this by:

- (i) limiting exploration depth.
- (ii) forcing systematic expansion.

→ IDS avoids being trapped in deep dead ends

part(c) - partial search tree (upto depth 4):



ms

limit: 0 → S

L=1 S → A → B → C

L=2 S → A → D → E → B → C → F → G

L=3 S → A → D → H (Exit-funnel)

ms

Name: Yash Raj

Roll no: 24K-073

Question #6.

pr=

part(a) & part(b) attached as .py file in zip folder.

part(c) :-

• Behaviour of Depth limited Dfs:

- ↳ Explores one branch deeply before exploring others
- ↳ if the depth limit is too small $\rightarrow$ goal not found.
- ↳ if limit is too large $\rightarrow$ may waste time exploring deep dead ends.
- ↳ Not guaranteed to find the shallowest solution.
- ↳ sensitive to depth-limit choice

In this maze:-

There're dead ends.

if Dfs explores wrong branch deeply, it wastes effort.
if depth-limit is smaller than actual goal depth, it fails.

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Behaviour of IDS

- ↳ Run Dfs repeatedly with increasing depth
- ↳ Explores level by level.
- ↳ Combines advantages of:
Dfs (low-memory)
Bfs (Completeness).

↳ Here IDS is more suitable because:-

- (i) It guarantees finding the goal if it exists.
- (ii) It doesn't require prior knowledge of goal depth.
- (iii) It uses memory efficiently.
- (iv) It avoids getting stuck in deep dead-ends.
- (v) It finds the shortest path in terms of depth.

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